Week 3 – RFM Customer Segmentation Analysis

Sub task 1

RFM Analysis is a customer segmentation technique used to analyze customer purchasing behavior based on:

- Recency
- Frequency
- Monetary Value

It helps businesses identify valuable customers, loyal customers, inactive customers, and customers who may stop purchasing.

This analysis was performed on the Superstore dataset.

Objective of the Project

The main objective of this project is to:

- Analyze customer purchasing behavior
- Identify high-value customers
- Detect at-risk and lost customers
- Improve customer retention strategies
- Support business decision-making through customer segmentation

Understanding RFM Metrics

Recency (R)

Recency measures how recently a customer made a purchase.

Business Meaning:

- Lower recency value is better
- Customers who purchased recently are more likely to purchase again
- Helps identify active customers

Frequency (F)

Frequency measures how often a customer makes purchases.

Business Meaning:

- Higher frequency indicates loyal customers
- Frequent customers contribute stable revenue
- Helps analyze customer engagement

Monetary Value (M)

Monetary value measures the total amount spent by a customer.

Business Meaning:

- Higher monetary value indicates high-value customers
- High spenders contribute more profit to the business
- Helps identify profitable customers

Dataset Used

Dataset Name:
Superstore Dataset

Important attributes used in the analysis:

- Customer ID
- Order Date
- Order ID
- Sales

Process Followed in the Project

Step 1 – Data Preparation

The dataset was loaded and checked for required customer and sales information. Date columns were converted into proper datetime format for accurate calculations.

Step 2 – Selecting Reference Date

A reference date was selected using the latest order date available in the dataset. This date was used to calculate recency for every customer.

Step 3 – Calculating RFM Metrics

Customer-wise aggregation was performed to calculate:

- Recency: Number of days since last purchase
- Frequency: Total number of transactions
- Monetary Value: Total amount spent by customers

Step 4 – Assigning RFM Scores

Customers were assigned scores from 1 to 5 for Recency, Frequency, and Monetary Value.

- Higher scores indicate better customers
- Recent customers received higher recency scores
- Frequent buyers received higher frequency scores
- High spenders received higher monetary scores

Step 5 – Customer Segmentation

Customers were classified into different segments based on their RFM scores.

Customer segments included:

- High Value Customers
- Loyal Customers
- At Risk Customers
- Lost Customers
- Other Customers

Customer Segment Explanation

High Value Customers

Characteristics:

- Recent purchases
- Frequent purchases
- High spending

Business Strategy:

- Loyalty rewards
- Premium offers
- Personalized services

Loyal Customers

Characteristics:

- Purchase frequently
- Regular engagement with the business

Business Strategy:

- Membership benefits
- Upselling opportunities
- Retention programs

At Risk Customers

Characteristics:

- Previously active customers
- Not purchasing recently

Business Strategy:

- Re-engagement campaigns
- Discounts and reminders
- Personalized offers

Lost Customers

Characteristics:

- Long inactive period
- Low engagement

Business Strategy:

- Win-back campaigns
- Special offers
- Customer feedback analysis

Advantages of RFM Analysis

- Simple and effective customer segmentation method
- Helps understand customer behavior
- Improves marketing efficiency
- Supports business decision-making
- Identifies profitable customers
- Enhances customer retention strategies

Business Impact of the Project

This project helps businesses:

- Identify valuable customers
- Improve customer retention
- Increase revenue through targeted marketing
- Reduce customer churn
- Build personalized customer engagement strategies
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Subtask 2: Customer Segmentation Based on RFM Scores

Introduction

Customer segmentation is the process of dividing customers into different groups based on their purchasing behavior. In this project, customers were segmented using RFM scores derived from Recency, Frequency, and Monetary analysis.

The purpose of customer segmentation is to help businesses understand customer behavior and create targeted marketing strategies.

Customer Segments Defined in the Project

1. High Value Customers

Characteristics:

- Recent purchases
- Frequent purchases
- High spending behavior

Business Importance:

- Generate maximum revenue
- Strong customer loyalty
- High engagement with the business

Recommended Strategy:

- Loyalty rewards
- Premium membership
- Exclusive offers

2. Loyal Customers

Characteristics:

- Frequent purchases
- Consistent customer engagement
- Moderate to high spending

Business Importance:

- Provide stable revenue
- Maintain long-term relationship with the business

Recommended Strategy:

- Personalized recommendations
- Membership benefits
- Upselling opportunities

3. At Risk Customers

Characteristics:

- Previously active customers
- Reduced recent activity
- Declining engagement

Business Importance:

- High probability of customer churn
- Possible reduction in future revenue

Recommended Strategy:

- Reminder emails
- Re-engagement campaigns
- Discount offers

4. New Customers

Characteristics:

- Recent purchases
- Low frequency of transactions
- Limited spending history

Business Importance:

- Potential future loyal customers
- Important for business growth

Recommended Strategy:

- Welcome campaigns
- Personalized onboarding
- Introductory offers

5. Lost Customers

Characteristics:

- Long inactive period
- Low engagement

- Low spending behavior

Business Importance:

- Minimal contribution to current revenue
- Indication of customer churn

Recommended Strategy:

- Win-back campaigns
- Customer feedback collection
- Special discount

6. Potential Loyalists

Characteristics:

- Moderate to high purchase frequency
- Good spending behavior
- Need better engagement

Business Importance:

- Can become loyal customers with proper retention strategies

Recommended Strategy:

- Targeted marketing
- Loyalty programs
- Personalized offers

Customer Segmentation Process

The customer segmentation process included the following steps:

1. Calculate RFM scores for each customer
2. Define customer segment rules
3. Assign customers to segments
4. Analyze customer behavior
5. Visualize customer distribution
6. Provide business recommendations

7. Summarize key insights

Customer Behavior Analysis

The analysis helped identify:

- Customers generating the highest revenue
- Customers purchasing frequently
- Customers becoming inactive
- Customers requiring retention strategies
- Customer groups with future growth potential

The segmentation process provided a better understanding of customer engagement and profitability.

Visualization of Customer Segments

Customer segment distributions were visualized using:

- Bar charts
- Pie charts

These visualizations helped understand:

- Relative size of each customer segment
- Most profitable customer groups
- Segments requiring business attention

Marketing Recommendations

High Value Customers

- Loyalty rewards
- Premium services
- Exclusive product access

Loyal Customers

- Membership programs

- Personalized recommendations
- Upselling strategies

At Risk Customers

- Re-engagement emails
- Discount campaigns
- Customer retention offers

New Customers

- Welcome offers
- Product recommendations
- Engagement campaigns

Lost Customers

- Win-back campaigns
- Customer surveys
- Special discounts

Key Insights

- High-value customers contribute significantly to business revenue.
- Loyal customers help maintain stable sales performance.
- At-risk customers require immediate retention strategies.
- Lost customers indicate customer churn issues.
- New customers represent future business growth opportunities.
- Customer segmentation improves personalized marketing and customer relationship management.

Business Impact

This customer segmentation analysis helps businesses:

- Improve customer retention
- Increase profitability

- Enhance marketing effectiveness
- Reduce customer churn
- Build better customer relationships
- Support data-driven business decisions

retention, improves profitability, and enhances overall business performance.

Week 3 – Subtask 3: Visualization of Customer Segments

1. Bar Chart for Customer Segment Distribution

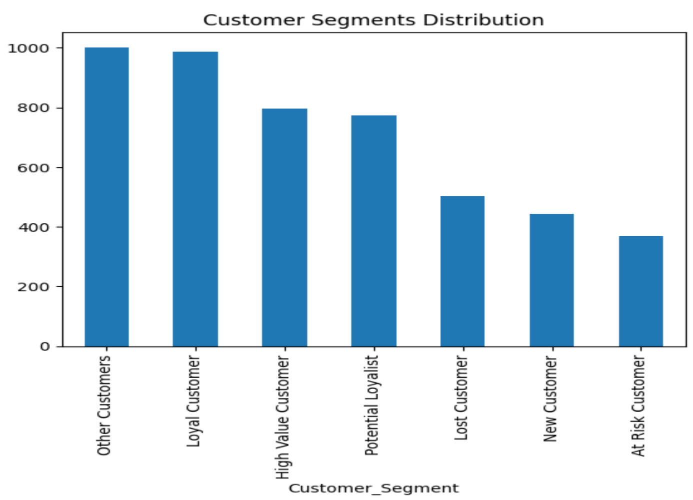
A bar chart was created to show the number of customers present in each segment.

Purpose:

- Compare customer segment sizes
- Identify dominant customer groups
- Detect risky customer categories

Business Insights:

- A large loyal customer segment indicates stable business performance.
- A high number of at-risk or lost customers indicates customer retention issues.
- A strong high-value customer segment indicates good profitability.



2. Pie Chart for Segment Proportions

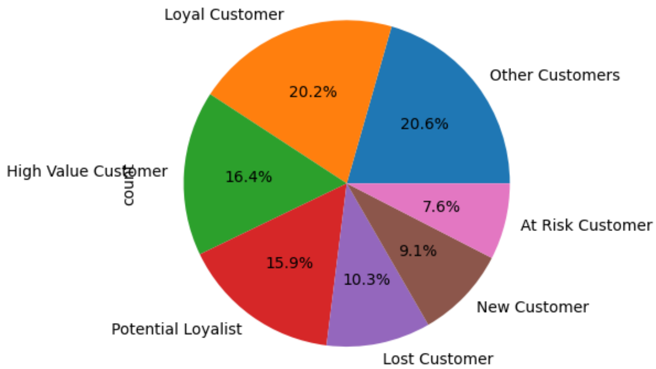
A pie chart was created to represent the percentage contribution of each customer segment.

Purpose:

- Understand proportional distribution of customer groups
- Analyze customer base composition
- Identify major contributing segments

Business Insights:

- High percentage of loyal customers reflects strong customer engagement.
- Large lost customer proportion indicates possible churn problems.
- Small high-value customer percentage suggests need for premium customer retention strategies.



3. Scatter Plot for RFM Relationship Analysis

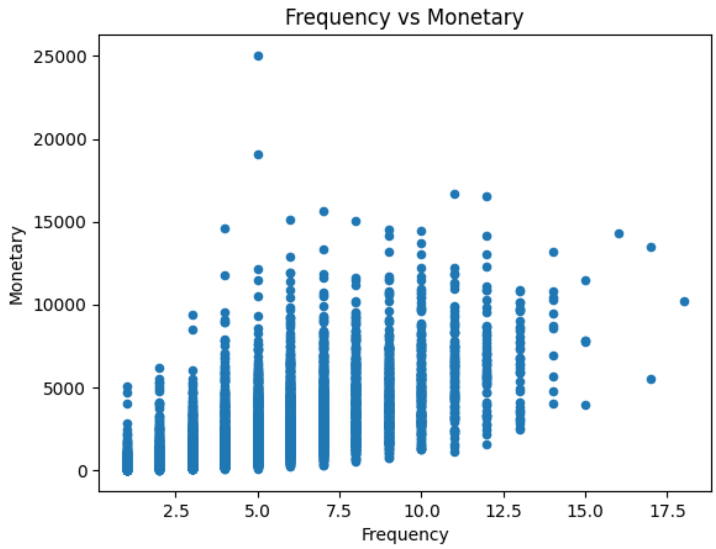
Scatter plots were used to visualize the relationship between Frequency and Monetary values.

Purpose:

- Identify customer clusters
- Analyze spending behavior
- Detect valuable customers
- Study engagement patterns

Business Insights:

- Customers with high frequency and high monetary value represent highly valuable customers.
- Customers with low frequency and low spending indicate weak engagement.
- Customers with high spending but low frequency may represent occasional premium buyers



4. Heatmap for RFM Distribution

A heatmap was created to analyze customer density across different RFM score combinations.

Purpose:

- Understand concentration of customer groups
- Detect high-value and low-value customer patterns
- Analyze customer retention behavior

Business Insights:

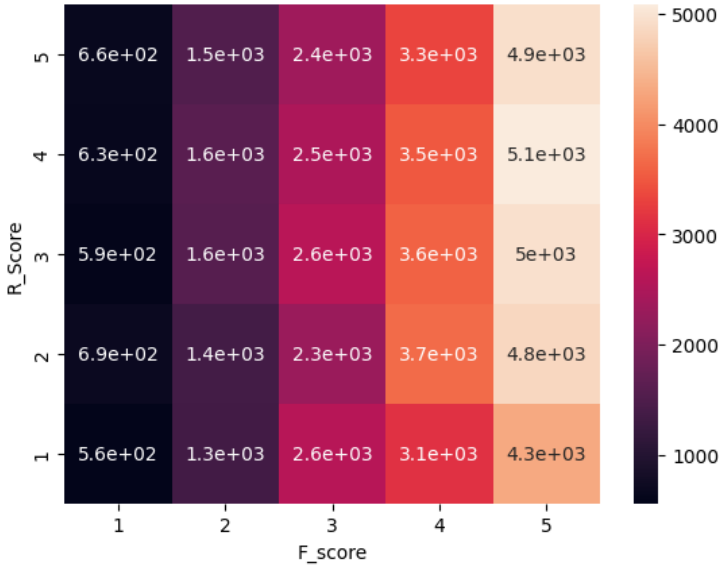
- Darker regions in higher RFM scores indicate strong customer engagement and profitability.
- Darker regions in lower RFM scores indicate inactive or weak customer groups.
- Heatmaps help businesses identify customer groups requiring targeted strategies.

Interpretation of Visualizations

The visualizations collectively helped identify:

- Most profitable customer groups
- Customers requiring re-engagement
- Customer purchasing behavior patterns
- Revenue-generating segments
- Potential customer churn risks

The charts provided a clear understanding of customer engagement and business performance.



Visualization of customer segments using RFM analysis provides meaningful business insights regarding customer engagement, spending behavior, and retention patterns. Charts such as bar charts, pie charts, scatter plots, and heatmaps help businesses identify profitable customers, understand customer distribution, and create targeted marketing strategies. These visualizations support effective customer relationship management and contribute to overall business growth